

Advanced (Carolina Reaper)

- Q1.** A music store sells x guitars for every 3 keyboards. If they sell 12 keyboards, how many guitars are sold?
(A) 4
(B) 6
(C) 8
(D) 9
(E) 12
- Q2.** If 5 ounces of coffee costs \$2.50, how much will 15 ounces cost?
(A) \$7.50
(B) \$10.00
(C) \$12.50
(D) \$15.00
(E) \$20.00
- Q3.** In a recipe, 2 cups of flour are needed for every $\frac{1}{4}$ cup of sugar. If 1 cup of sugar is used, how many cups of flour are needed?
(A) 8
(B) 10
(C) 12
(D) 14
(E) 16
- Q4.** A car travels 250 miles in 5 hours. How many miles will it travel in 10 hours?
(A) 250
(B) 375
(C) 500
(D) 750
(E) 1000
- Q5.** A group of friends want to split some money in the ratio 2 : 3 : 5. If they have a total of \$200 to split, how much will the person with the ratio 5 get?
(A) \$20
(B) \$30
(C) \$40
(D) \$50
(E) \$100
- Q6.** A bakery sells 480 loaves of bread per day. They pack the loaves in bags of 12. How many bags will they need per day?
(A) 30
(B) 40
(C) 50
(D) 60
(E) 80
- Q7.** A water tank can hold 3000 gallons of water. If 1200 gallons of water are already in the tank, what percentage of the tank is filled?
(A) 20%
(B) 30%
(C) 40%
(D) 50%
(E) 60%
- Q8.** A plane travels 600 miles in 3 hours. How many miles will it travel in 6 hours?
(A) 1000
(B) 1200
(C) 1500
(D) 1800
(E) 2000

Open Ended Questions (FRQ)

- Q9.** A school has 300 students. The ratio of boys to girls is 7 : 5. How many boys and girls are in the school? Show your work and explain your reasoning.
- Q10.** A company has 250 employees. The company wants to give a 10% raise to all employees. If the total budget for raises is \$15000, will the company have enough money to give all employees the raise? Show your work and explain your reasoning.

Chef's Tips

- Tip 1: Make sure students show their work for all problems
- Tip 2: Emphasize the importance of using proportions to model real-world situations
- Tip 3: Encourage students to check their units and make sure they are consistent